SmartPCB-XAI

Intelligent Optical Inspection & Analytics Platform (Industry 4.0)

INSPECTION CERTIFICATE

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Inspection ID:	INSP-1790170469	**AI Inspection Score:**	**52.2 / 100**
Image ID:	IMG-3195c860e31c	**Pipeline Status:**	**FAIL**
Generated By:	Automated AI Inspection Engine (no human inspector)	**Score Grade:**	REJECT
Model Version:	component_best.pt + defect_best.pt	**Defect Count:**	5

1. Detected Manufacturing Defects & Severity Ratings

#	Defect Type	Confidence	Severity Level	Severity Score	Impacted Area / Proximity
1	copper	90.6%	MEDIUM	35.1/100	not resolved
2	pin-hole	89.3%	HIGH	60.0/100	not resolved
3	mousebite	85.9%	HIGH	54.4/100	not resolved
4	spur	83.5%	MEDIUM	44.1/100	not resolved
5	open	82.8%	CRITICAL	84.9/100	not resolved

2. PCB Component Verification Summary

Component Family	Detected Count	Detection Status	Note
No discrete components detected	0	n/a	See pipeline status above

3. Rule-Based Root-Cause Guidance & Suggested Engineering Actions

Target Defect	Probable Root Cause (not proven)	Suggested Engineering Action	Reference Guidance
Spurious Copper Residue	• Contaminated etching solution containing suspended copper particulate. • Incomplete wash/rinse cycle post-etching.	• Exposed copper flake: remove gently with fine precision tweezers or micro-vacuum. • Embedded particle: route out non-conductive area and re-seal with UV resin.	IPC-A-610G Section 10.5 (Extraneous Metallic Residue)
Pin-Hole / Via Void	• Air entrapment or outgassing during PTH (Plated Through Hole) copper plating. • Moisture absorption in FR4 core before thermal lamination.	• Fill surface pinhole with electronic-grade conductive silver epoxy or reflow solder. • Smooth flush with pad surface using thermal flat blade.	IPC-A-610G Section 9.1 (Plated-Through Hole Voids)

Mousebite / Trace Edge Erosion	• Uneven chemical etching or particulate contamination in acid bath. • Defective photoresist film adhesion along high-density signal tracks.	• If remaining trace width > 80% nominal: clean and apply conformal overcoat to prevent oxidation. • If remaining trace width < 80%: reinforce by soldering tinned copper ribbon jumper per IPC-7721.	IPC-A-610G Section 10.3 (Conductor Width Reduction)
Copper Spur / Protrusion	• Incomplete chemical etching leaving residual copper spur. • Micro-crack in photolithography master artwork (photomask).	• Use stereoscopic surgical micro-scalpel to excise unwanted copper protrusion. • Clean debris with vacuum wand and apply insulating lacquer.	IPC-A-610G Section 10.4 (Minimum Conductor Spacing)
Trace Disconnection / Open Circuit	• Etching over-exposure during inner/outer layer PCB chemical fabrication. • Mechanical scratching or abrasion during manual handling / depaneling.	• Carefully scrape solder mask 2mm on either side of the open break using micro-chisel. • Tin exposed copper trace using fine chisel soldering tip.	IPC-7711/7721 Procedure 4.2.1 (Conductor Repair Using Wire Jumper)

4. XAI Attribution — Interpretability Summary

Attribution Method	activation_energy_approximation
Scientific Status	APPROXIMATION (activation-energy, NOT true gradient Grad-CAM++)
Disclaimer	Not an authentic gradient-based Grad-CAM++. This visualization is a feature-map energy approximation (mean absolute channel activation) for qualitative interpretability and must not be cited as a gradient attribution proof. Part 2 will integrate true gradient-weighted Grad-CAM++.
Explained Detection	open (D-5)
Engine Interpretation	Qualitative activation-energy heatmap highlights bottleneck feature regions near 'open' detection. This is a visualization heuristic, not a gradient proof. See DISCLAIMER field.
Interpretation Guidance	Hot regions (red/yellow) in XAI overlay indicate regions of high feature-map activation associated with the model's
XAI Heatmap Shape	not reported
Overlay Image Present	YES

5. Inference Timing & Model Metadata

Stage	Mean Time (ms, if available)
0. Preprocessing (CLAHE/resize)	33.2308
1. Component YOLO Inference	1450.235
2. Defect YOLO Inference	2340.7773
3. XAI Attribution (approx)	2506.6985
4. Postprocess / Severity / KB	7.3346
Total Inspection Round-Trip	6338.4129
Model / Engine Field	**Value**
Component Model File	component_best.pt
Defect Model File	defect_best.pt
Engine Version	1.0.0-phase1
Inference Device	cpu
CUDA Available (actual measured)	not reported
Component Model Class-Count Match to canonical 6?	See /health endpoint model_status.component_model.verification.class_names_match
Defect Model Class-Count Match to canonical 6?	See /health endpoint model_status.defect_model.verification.class_names_match

6. Confidence Distribution — Detection Quality

Confidence Bin	Detection Count	Share (%)
Very Low (0.00 ≤ c < 0.30)	0	0.0%
Low (0.30 ≤ c < 0.50)	0	0.0%
Medium (0.50 ≤ c < 0.70)	0	0.0%
High (0.70 ≤ c < 0.90)	4	80.0%
Very High (0.90 ≤ c ≤ 1.00)	1	20.0%
Total Detections	**5**	**100.0%**

7. Limitations & Scientific Disclaimers

7.1 XAI Approximation: The XAI overlay is an activation-energy approximation (mean absolute channel feature-map energy). It is **NOT an authentic gradient-weighted Grad-CAM++** and must not be cited as causal gradient evidence.

7.2 Severity Score is Project-Defined: All severity scores (5–100) and levels (LOW/MEDIUM/HIGH/CRITICAL) are computed using the **SmartPCB-XAI project-defined deterministic formula** documented in ai/severity/severity_engine.py. They are not an IPC-A-610 numeric score. IPC-A-610 is used solely as qualitative workmanship guidance reference text in the repair KB.

7.3 Root-Cause = Rule-Based Recommendation Engine, Not Learned Causal Model: Probable causes and repair actions are outputs of a **Rule-Based Root-Cause Recommendation Engine** (see ai/recommendation/repair_kb.py). They are framed as *possible causes*, not proven causes. No causal inference ML model has been trained or validated.

7.4 mAP >96% is Aspirational Only: Historical claims of 97.6% component mAP or 98.2% defect mAP are **UNVERIFIED** in this audit. The only reproducible training artifacts are partial epoch CSVs (epoch-1 component mAP50=0.018, epoch-2 defect mAP50=0.456). Full training-to-convergence with a held-out leakage-free test set has not been reproduced.

7.5 Board-Level Data Leakage = PARTIALLY VERIFIED: Cross-split board-ID and filename intersection = 0 for both datasets (see DATASET_REPORT.md — NEW UPDATE 2026-09-23). Perceptual-hash (pHash) near-duplicate cross-split detection is UNAVAILABLE because the imagehash pip package is not installed; 0.0% leakage in the perceptual sense remains UNVERIFIED.

7.6 CPU-Only Benchmarked Environment: Baseline inference benchmark measured on CPU (torch 2.13.0+cpu, no CUDA). Synthetic 640×640 image mean total = 2,508 ms (N=3). Historical claim of 16.1 ms latency is UNVERIFIED — that figure would require a GPU (A100/RTX) with batch optimization, FP16, and/or TensorRT compilation, none of which have been validated in this audit environment.

7.7 Class Taxonomy: Two Different Vocabularies Coexist (VERIFIED 2026-09-23): The deployed component checkpoint emits 10 classes (relay, buzzer, ic, switch, battery, transistor, capacitor, resistor, led, diode); the curated component dataset defines a different 6-class vocabulary. The deployed defect checkpoint emits the original DeepPCB names (missing_hole, mouse_bite, open_circuit, short, spur, spurious_copper) which are alias-normalized to the curated 6 (open, short, mousebite, spur, copper, pin-hole). Both checkpoints PASS the model-safety checkpoint verification (exist / load / class-count / device) with alias-aware name mapping (see /health model_status). COMPONENT_MAPPING_STATUS and DEFECT_MAPPING_STATUS are both **PARTIAL**: the mapping is name-based and lossy in one direction — 'inductors' exists in the curated component taxonomy but no deployed class can emit it, and 'missing_hole' is only an approximate alias of 'pin-hole'. Read every per-class table with that asymmetry in mind.

7.8 Class Imbalance Warning: Component dataset: inductors = 620 / 33,156 total boxes (~1.87%). Defect dataset: copper = 1,474 / 10,013 (~14.7%). Minority classes may have depressed precision/recall even at full convergence. Class-balanced loss or focal loss may be warranted at retraining time.

7.9 Reports Contain No Fake Metrics: PDF / Excel reports only contain numbers actually computed from inspection_data or model safety endpoints. No placeholder performance figures (e.g. 97.6%, 16.1ms, 0.0%) are embedded into generated reports unless they have been genuinely produced by a validated run in this audit.